

Atlas Platform Technical Overview

Atlas Platform is a web-based Geographic Information System (GIS) platform designed for spatial data analysis and visualization.

The platform supports uploading raster datasets in GeoTIFF format, which are automatically converted to Cloud Optimized GeoTIFFs (COG) for efficient tile serving.

Vector datasets in GeoPackage and GeoJSON formats are also supported. Each dataset can be used to create map layers with custom styles.

The AI chat agent can perform geospatial analysis, run Python scripts in a sandboxed environment, and answer questions about uploaded documents.

Document RAG (Retrieval-Augmented Generation) allows users to upload PDF documents and query their contents through the chat interface.

Embeddings are generated using OpenAI text-embedding-3-small model and stored in pgvector for cosine similarity search.

The platform uses a signal-driven architecture where model changes publish events to Redis, which are delivered to clients via Server-Sent Events.

Celery workers handle heavy processing tasks including COG generation from raster uploads and PDF document processing for RAG.